
BirdClef 2026: Solutions using Custom CNN Architectures

September 10, 2026

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Abstract

This project addresses the Sound Event Detection task (BirdCLEF 2026) by proposing two custom CNNs trained from scratch on Mel-spectrograms. The approach is evaluated, both individually and ensembled, against a pre-trained baseline. Furthermore, a critical study on the impact of spatial Data Leakage in data splitting strategies is presented.

1. Introduction

The BirdCLEF+ 2026 challenge addresses the Sound Event Detection (SED) task, complicated by overlapping signals, severe class imbalance, and stationary noise.

This project develops an acoustic pipeline from scratch rather than relying on massive pre-trained networks. Audio signals are processed using Mel-spectrograms and Per-Channel Energy Normalization (PCEN) for noise suppression. Two custom Convolutional Neural Networks (CNNs) were developed from scratch: an Asymmetric CNN and an SE-ResNet. Their effectiveness is evaluated against a pre-trained baseline.

The project faces the issue of Data Leakage by showing how a random split allows models to memorize background noise, improving competition metrics at the expense of true generalization capability in novel environments.

2. Related Work

Deep learning has outperformed traditional methods in audio classification. The literature highlights the combination of Mel-spectrograms and Convolutional Neural Networks (CNNs) as the most effective approach in this domain. However, deep models trained on limited data are prone to overfitting, an obstacle frequently mitigated through *Data Augmentation*. Furthermore, since the time and frequency

axes are not homogeneous like natural images, research explores the use of asymmetric convolutional filters. Finally, studies warn that performance is heavily compromised by noisy or biased data (Zaman et al., 2023; Purwins et al., 2019).

3. Method

3.1. Acoustic Pre-processing

Exploratory Data Analysis (EDA) revealed severe class imbalance in isolated tracks and high polyphony in soundscapes (Figure 1), with frequent secondary labels.

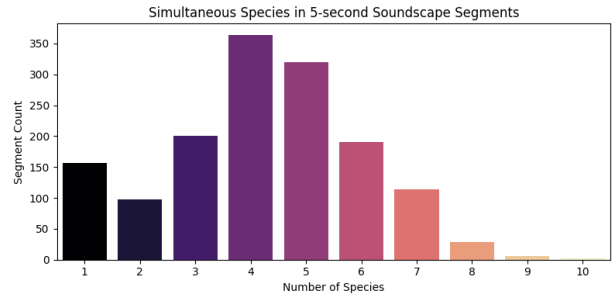


Figure 1. Distribution of simultaneous bird species within 5-second soundscape segments.

The audio signals (32 kHz) were fragmented into non-overlapping 5-second segments, selecting those with the maximum Root Mean Square (RMS) energy. Converted into high-resolution Mel-spectrograms, *Per-Channel Energy Normalization* (PCEN) was applied instead of the traditional logarithmic conversion. PCEN mathematically suppresses stationary background noise, effectively enhancing the bird calls (Figure 2).

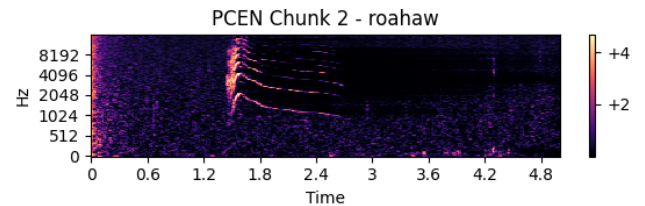


Figure 2. A 5-second PCEN spectrogram of a bird call.

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Deep Learning and Applied AI 2026, Sapienza University of Rome, 2nd semester a.y. 2025/2026.

3.2. Architectures and Training

To mitigate imbalance, classes were capped at 150 samples, and soft targets were assigned confidences of 1.0 and 0.3 for primary and secondary species, respectively.

Two proprietary architectures were implemented:

- **Asymmetric CNN:** Replaces traditional square kernels with asymmetric filters (1×5 and 5×1) to exploit time-frequency asymmetry and reduce parameters.
- **SE-ResNet:** A residual architecture with *Squeeze-and-Excitation* blocks to dynamically recalibrate the importance of the extracted channels.

Both networks culminate in an *Attentive Pooling* layer, allowing the model to focus exclusively on active frames to produce clip-level predictions. Robustness is ensured by a composite *Data Augmentation* pipeline: Gaussian noise and random temporal shifts on the waveforms, and SpecAugment (time-frequency masking) on the spectrograms.

3.3. Two-Phase Training Strategy

To bridge the domain gap between isolated recordings and real-world environments, the training process was structured in two phases. Phase A consists of pre-training on focal tracks, enabling the networks to learn the primary species features. Subsequently, Phase B performs fine-tuning on soundscapes, adapting the weights to handle high polyphony and background noise.

To counteract class imbalance, Binary Cross-Entropy was replaced by a multilabel Focal Loss ($\gamma = 2.0$, $\alpha = 1.0$), which dynamically down-weights well-classified species to focus the gradient on rare ones. Additionally, Phase A applied Mixup augmentation by linearly blending waveforms and combining labels via (`torch.max`) to preserve soft targets. The pipeline was optimized using AdamW and a OneCycleLR scheduler, monitoring validation performance and model selection directly through the validation Focal Loss. Finally, during the inference phase, Test-Time Augmentation (TTA) and temporal prediction smoothing were applied. This post-processing strategy stabilizes the outputs, reducing false positives and improving overall robustness on continuous audio streams.

3.4. Critical Analysis: Data Leakage

The initial pipeline used a random stratified split and duplicated singleton species. However, this approach introduces severe spatial Data Leakage. By fragmenting continuous recordings into 5-second segments and shuffling them randomly, adjacent fragments from the same file appear in both the training and validation sets. Consequently, models

overfit the background noise (e.g., microphone characteristics) rather than learning the actual bird calls.

To quantify this environmental bias, an ablation study was conducted using a purified variant of the Asymmetric CNN. The random split was replaced by a Group Split based on the file identifier, ensuring segments from the same track remained in a single set. Singletons were strictly isolated in the training set, and a 500 Hz high-pass filter was applied to suppress low-frequency noise.

As expected, purging the leakage caused a natural score decline. This demonstrates how standard random splits can artificially inflate evaluation metrics through environmental bias, failing to reflect true generalization in unseen domains.

Code. The source code used for feature extraction, model training, and ensemble submissions is publicly available at: https://github.com/AdriSemerano/DLAI_2026_Project_Semerano.git.

4. Results

Table 1 summarizes the public leaderboard performance. While a pre-trained ensemble (ResNet34d + EfficientNet-B0) peaked at 0.850, our custom ensemble achieved a competitive 0.835, outperforming large pre-trained baselines (0.818). This confirms that lightweight architectures designed for time-frequency representations, combined with PCEN filtering, effectively extract robust features.

Furthermore, the purified Asymmetric CNN yields a score of 0.786 (~ 0.034 gap), proving that naive random splits overestimate true acoustic generalization.

Table 1. Public Leaderboard performance comparison.

Model / Ensemble	Split	ROC-AUC
Pre-trained Ens. (ResNet34d + EffNetB0)	Random	0.850
Custom Ens. (Asym. + SE-ResNet)	Random	0.835
Asymmetric CNN	Random	0.820
Pre-trained Ens. (EffNetB2 + ConvNeXt)	Random	0.818
SE-ResNet	Random	0.815
Asymmetric CNN (Purified)	Group	0.786

5. Conclusions

Targeted architectural designs (asymmetric kernels, PCEN, Squeeze-and-Excitation) rival massive pre-trained networks. Crucially, addressing spatial leakage via rigorous Group Splits prevents validation inflation, trading short-term metrics for genuine real-world generalization. Future work will incorporate pseudo-labeling while preserving validation purity.

6. Statement on the use of AI

During the preparation of this project, AI tools were utilized to support some phases of the workflow. Specifically, AI assistance was employed for:

- **Research exploration:** Assisting in exploring papers and reports in literature.
- **Code assistance and debugging:** Helping troubleshoot implementation errors, assisting with code writing, and cleaning up scripts throughout pipeline development.
- **Text translation and synthesis:** Translating concepts into formal academic English and condensing descriptions to meet the strict two-page report limit.

References

- Purwins, H., Li, B., Virtanen, T., Schlüter, J., Chang, S., and Sainath, T. Deep learning for audio signal processing. *Journal of Selected Topics of Signal Processing*, 13 (2):206–219, 2019.
- Zaman, K., Sah, M., Direkoglu, C., and Unoki, M. A survey of audio classification using deep learning. *IEEE Access*, 2023. doi: 10.1109/ACCESS.2023.3318015.

Appendix: Exploratory Architectures

During the preliminary exploration phase, alternative architectures—specifically Convolutional Recurrent Neural Networks (CRNNs) and Multi-Scale feature extraction models—were implemented and evaluated. These configurations yielded inferior performance, achieving a Private Leaderboard ROC-AUC score of approximately 0.75. Due to their limited capacity to handle the high variability and noise of the soundscapes compared to the final Asymmetric CNN and SE-ResNet, these exploratory models were ultimately discarded.